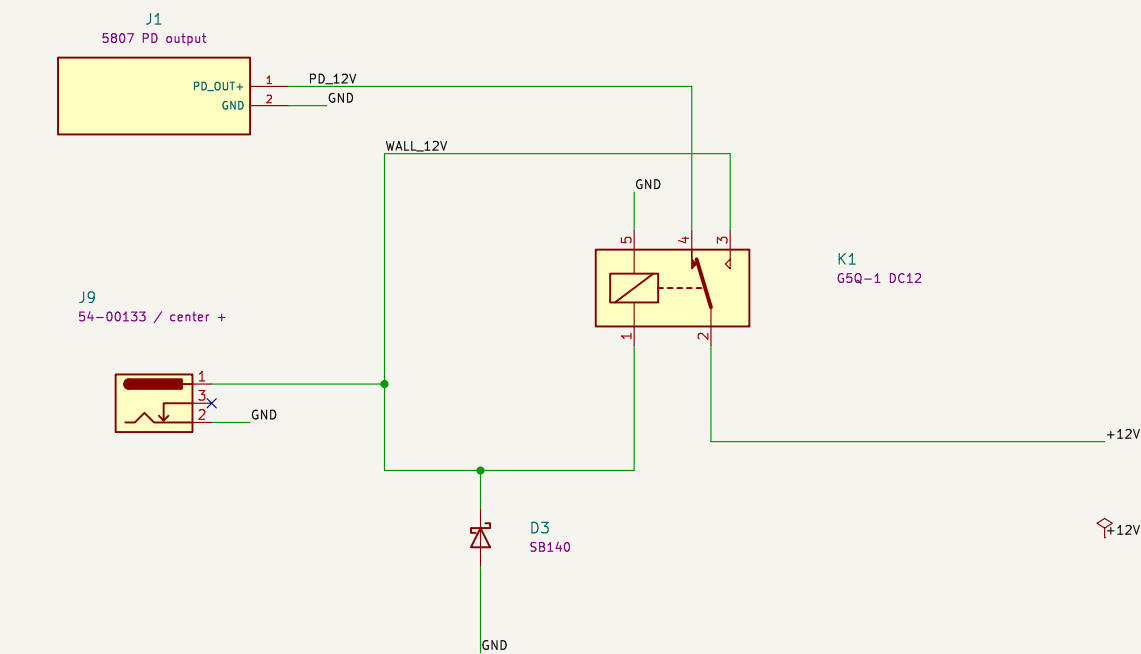
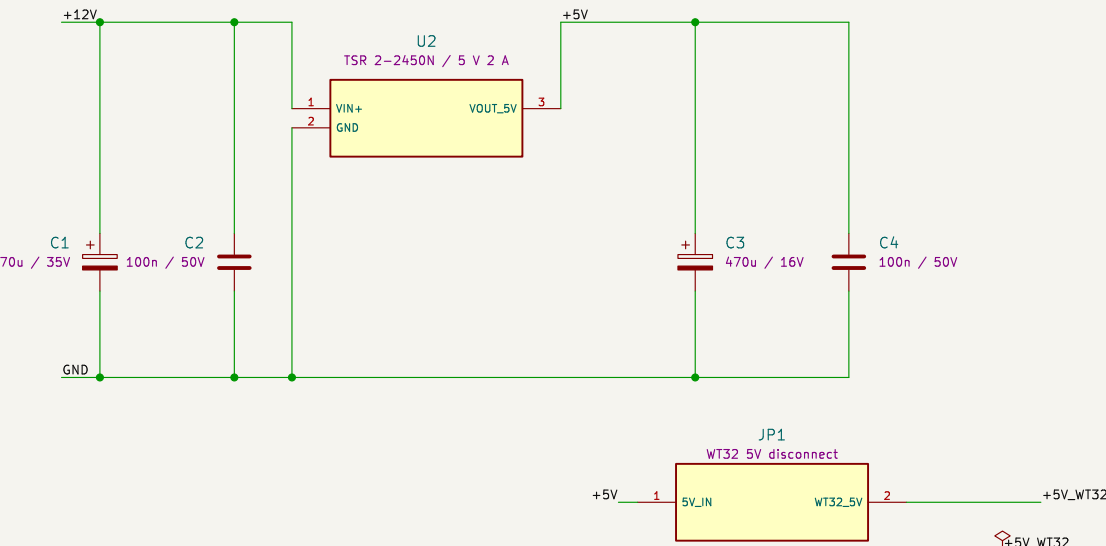


01 INPUT POWER / AUTOMATIC WALL PRIORITY



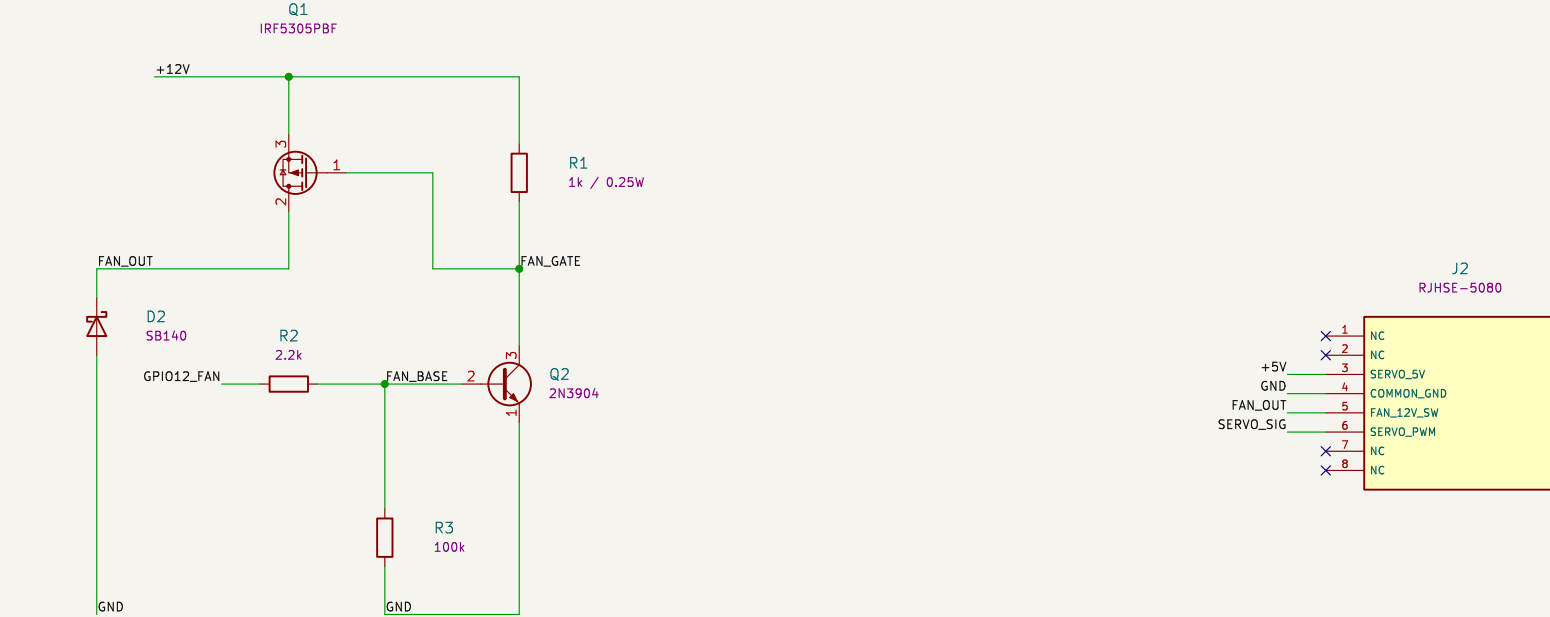
NC: USB PD NO: wall COM: selected +12V
12 V regulated / 3 A maximum sources; common GND.

02 FIXED 5 V REGULATOR / WT32 DISCONNECT



Place C2 and C4 at U2 pins. Remove JP1 before
powering the WT32 from its own programming USB.

03 HIGH-SIDE FAN / SERVO / POWERED RJ45

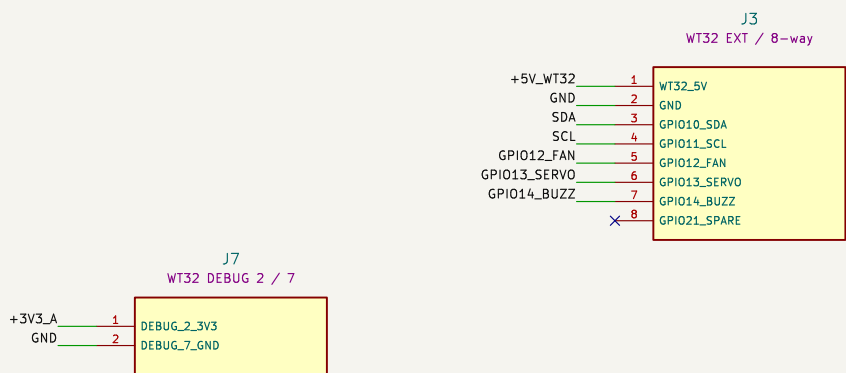


GPIO12 HIGH = fan ON; reset = OFF. Start PWM at 100 Hz.

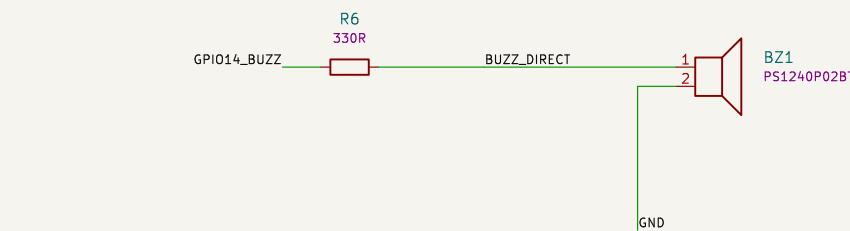
RJ45: 1.5 A/contact; pin 4 carries fan + servo return.

04 WT32 DISPLAY / CONTROL HARNESSSES

J3 is the EXT cable. J7 takes only DEBUG pins 2 and 7.
3.3 V analog excitation and ADC power come from J7.



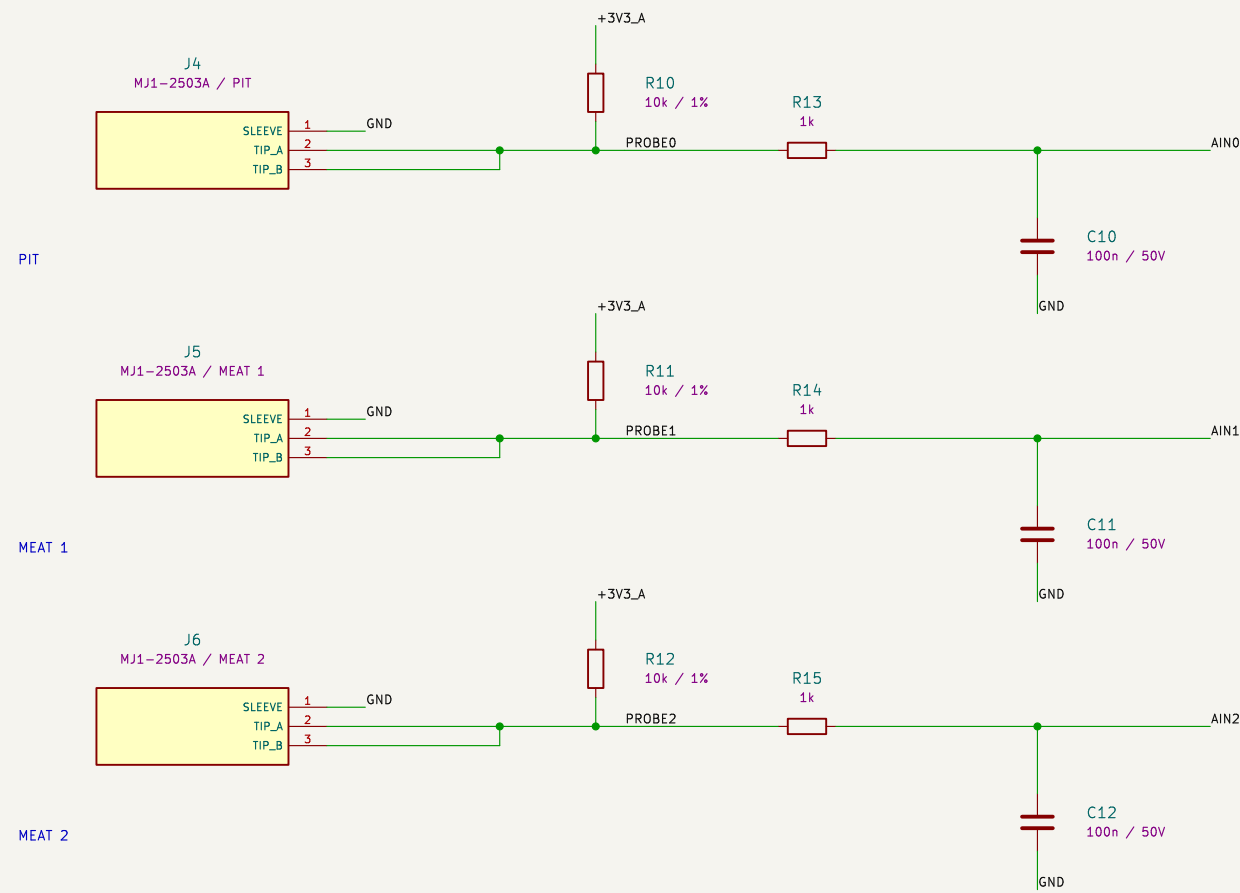
05 PASSIVE PIEZO



GPIO14 -> 330R -> passive piezo. Use about 4 kHz.

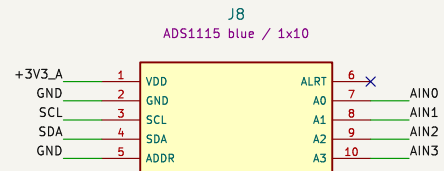
No DC buzzer or magnetic speaker substitution.

06 THREE NTC PROBES / FILTERS / SOCKETED ADS1115



AIN3 measures excitation through R16.

ADDR = GND (0x48). ALERT is unused.



NTC resistance = 10k * raw / (raw_supply - raw). Verify with known resistors before thermal calibration.

ASSEMBLY / OPERATING NOTES

POWER SOURCES

MOD1: Adafruit 5807, configured for 12 V / 3 A.

J9: 5.5 x 2.1 mm, regulated 12 V, center positive.

The J9 sleeve switch (pin 3) is unused.

K1: coil pins 1/5; COM 2, NO 3, NC 4.

D3 cathode is on raw WALL_12V; anode is on GND.

Switchover may restart the controller.

No onboard fuses: use protected/current-limited sources.

REGULATOR / ACTUATORS

U2 supplies 5 V / 2 A total for WT32 and servo.

RJ45: 1.5 A/contact; pin 4 is fan + servo return.

This connector is NOT Ethernet or PoE.

Confirm connector, cable and stall-current limits.

Q1: pin 1 gate; pin 2 drain/live tab; pin 3 source.

Q1 mounts flat on a 1 mm insulating support.

ADC / SENSORS

J8 is the owned 28 x 17 mm blue ADS1115 module.

Pin 1 = VDD; pin 10 = A3. Confirm module orientation.

Module must provide local bypass and I2C pull-ups to VDD.

Probe TIP_A and TIP_B contacts are intentionally joined.

BUILD STATUS

Electrical revision S4; native labels connect between blocks.

Module fit, load tests and thermal validation remain required.

See README for release checks before fabrication/assembly.

ENGINEERING PROTOTYPE – VERIFY MODULES, LOADS AND ASSEMBLY BEFORE RELEASE

Sheet: /
File: pitclaw-carrier.kicad_sch

Title: Pit Claw carrier / relay source selection

Size: A2 Date: 2026-09-06

Rev: S4

Id: 1/1